

Cybersecurity Threat Assessment and Mitigation Plan
ShopSecure E-Commerce Platform

A Documentation Report submitted to
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ITEP 311 - Information Assurance 1
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I. Introduction

ShopSecure is a fictional e-commerce company that handles online transactions and stores sensitive customer information such as personal details, login credentials, and payment data. Recently, the company has experienced **unauthorized access attempts, phishing emails, and unusual transactions**, raising serious cybersecurity concerns.

This documentation identifies potential **threats and vulnerabilities**, analyzes their **impact**, and proposes **mitigation strategies** to protect the website, business operations, and customer data.

II. Identification of Threats and Vulnerabilities

2.1 Technical Vulnerabilities

These are weaknesses in the system's technology, software, or network infrastructure.

- Weak password policies and lack of multi-factor authentication (MFA)
- Outdated software, plugins, or unpatched systems
- Insecure APIs and poorly configured servers
- SQL injection and cross-site scripting (XSS) vulnerabilities
- Lack of encryption for stored or transmitted data
- Inadequate firewall and intrusion prevention systems
- Poor session management

2.2 Human Factors

Human behavior often creates security gaps.

- Employees falling for phishing or social engineering attacks
- Weak or reused passwords by staff or customers
- Lack of cybersecurity awareness training
- Insider threats (malicious or careless employees)
- Improper handling of sensitive customer data
- Employees accessing systems beyond their job role

2.3 Physical Threats

Physical access to systems can bypass digital security.

- Unauthorized access to server rooms or office spaces
- Theft or loss of company laptops, USB drives, or backups
- Damage to infrastructure due to fire, flood, or power failure
- Poor surveillance or access control in office facilities

2.4 Operational Risks

Risks arising from internal processes and third-party dependencies.

- Poor incident response and recovery procedures
- Lack of regular security audits and risk assessments
- Dependence on third-party vendors with weak security
- Inadequate data backup and disaster recovery plans
- Weak transaction monitoring systems
- No clear cybersecurity policies or enforcement

III. Impact Analysis

3.1 Assets at Risk

- Customer personal data (names, addresses, phone numbers)
- Payment information (credit/debit card data)
- Login credentials and account data
- Company financial records
- Brand reputation and customer trust
- Business continuity and revenue

3.2 Potential Consequences of a Breach

If a security breach occurs, ShopSecure may face:

- Financial losses due to fraud and chargebacks
- Identity theft and financial harm to customers
- Legal penalties for data protection violations
- Loss of customer trust and reputational damage
- Website downtime and disrupted operations
- Increased cost for recovery and remediation

3.3 Exploitation of Vulnerabilities

- Hackers could exploit outdated software to gain system access
- Phishing emails could trick employees into revealing credentials
- Weak authentication could allow brute-force attacks
- Unsecured APIs could expose customer data
- Physical access could enable data theft or sabotage
- Poor monitoring could allow fraud to go unnoticed

IV. Mitigation Strategies

4.1 Preventive Measures

Actions to reduce the likelihood of attacks:

- Enforce strong password policies and multi-factor authentication
- Regularly update and patch systems and software
- Encrypt sensitive data at rest and in transit
- Implement role-based access control (RBAC)
- Conduct employee cybersecurity awareness training
- Secure server rooms with access control and surveillance
- Perform regular vulnerability assessments and penetration testing

4.2 Detection Strategies

Measures to identify threats early:

- Deploy intrusion detection and prevention systems (IDS/IPS)
- Monitor network traffic and system logs continuously
- Implement real-time fraud detection for transactions
- Use email filtering and anti-phishing tools
- Set up alerts for suspicious login attempts
- Conduct routine security audits

4.3 Response Plans

Steps to take once a security incident occurs:

- Establish a formal incident response plan
- Isolate affected systems immediately
- Notify affected customers and relevant authorities
- Reset compromised credentials
- Restore systems using secure backups
- Conduct post-incident analysis to prevent recurrence
- Update policies and controls based on lessons learned

Summary

5.1 Identified Threats and Vulnerabilities

- Technical flaws (weak authentication, outdated systems)
- Human errors (phishing, poor password practices)
- Physical risks (unauthorized access, device theft)
- Operational weaknesses (lack of monitoring and response plans)

5.2 Impact Overview

- Customer data theft and financial fraud
- Legal and regulatory consequences
- Damage to reputation and loss of trust
- Business disruption and financial losses

5.3 Proposed Mitigation Strategies

- Strengthen technical defenses and access controls
- Train employees on cybersecurity awareness
- Improve monitoring and detection systems
- Establish clear response and recovery procedures

Conclusion

Cybersecurity threats pose significant risks to ShopSecure's operations and customer trust. By identifying vulnerabilities, understanding their impact, and implementing strong preventive, detection, and response strategies, ShopSecure can significantly reduce the likelihood and severity of cyber incidents. A proactive and layered security approach is essential for protecting e-commerce platforms in today's digital environment.